ADVANCE TRAUMA

Statistical Analysis Plan

Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Version 0.6.0, 2026-03-04

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1 Administrative information

1.1 Study identifiers

- Protocol version 1.5.0 dated 2025-06-12
- ClinicalTrials.gov identifier: NCT06321419
- Clinical Trials Registry - India identifier: CTRI/2024/07/071336

1.2 Changelog

Once version 1.0.0 is finalised, this section will be updated with a changelog.

1.3 Contributors

Name and ORCID	Affiliation	Role
Martin Gerdin Wärnberg (MGW)	Karolinska Institutet	Principal Investigator
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1.4 Use of generative artificial intelligence

Generative artificial intelligence tools, including large language models accessed via Cursor, were used to assist with drafting, revising, and formatting this statistical analysis plan. All substantive statistical content was reviewed and approved by the authors, who take full responsibility for the final document.

2 Trial synopsis

Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and at least 4320 patients.

Eligibility Criteria

Clusters are secondary or tertiary hospitals providing trauma care in India that admit or refer/transfer for admission at least 400 patients with trauma per year. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained.

Participants are adult trauma patients who present to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach for collection of routinely recorded data. We will obtain informed consent for collection of non-routinely recorded data, such as quality of life and disability outcomes. Patients who are unconscious or lack a legally authorized representative will be included under a waiver of informed consent. Note that consent here refers to consent to data collection.

Trial Period February 2025 to October 2029

3 Special considerations

3.1 Funding

This trial is not yet fully funded. The Trial Management Group has decided to proceed with the trial with the expectation that additional funding will be secured. The Trial Steering Committee will be informed of the funding status at each meeting. If funding is not secured, the trial will be stopped. This will likely result in an underpowered trial. The justification for this decision is that the intervention is considered standard of care in many countries and the data collection is considered minimal risk. There is therefore a very small risk of harm to patient participants, but a potential direct benefit to those patient participants who receive the intervention. The benefit-risk ratio is therefore considered to be favourable, even in the case of an underpowered trial.

4 Aim and objectives

4.1 Aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

4.2 Objectives

- To estimate the effects of ATLS® training on 30-day in-hospital mortality compared to standard care.
- To assess the robustness of the main analysis results to different model specifications through sensitivity analyses.
- To estimate the effects of ATLS® training on secondary outcomes compared to standard care.
- To explore the effects of ATLS® training on subgroups of patients compared to standard care.

5 Statistical analysis

5.1 Design

This is a batched stepped-wedge cluster randomised trial, including a total of 30 clusters (see Figure 1). Each cluster is a hospital and represents the unit for randomisation, even if one or more units are trained in the same hospital. There are two intervention conditions: ATLS® training and standard care. The trial is designed and will be analysed within a superiority framework, with ATLS® training compared with standard care for the primary and secondary outcomes.

The trial will be composed of 6 batches of identical 12-period 5-sequence designs, with one cluster allocated to each sequence of each batch (equal allocation across sequences)[1]. Each period is one month, and each cluster will be in the trial for a total of 13 months. Clusters are repeatedly measured across periods. The intervention will be implemented during a one-month transition period, which will be excluded from the analysis. There will be an anticipated overlap of 6 months between successive batches.

Clusters will be sequentially allocated to batches based on when they enter the study, and within each batch clusters will then be randomised to an intervention implementation sequence. Individual patients will be included cross-sectionally for the primary outcome, although some secondary outcomes will be assessed at multiple follow-up time points. The anticipated trial period is from February 2025 to October 2029.

Because certain secondary outcomes will be more burdensome to collect and because we anticipate larger effects on these outcomes, we will nest a staircase design within the main stepped-wedge design to measure these outcomes (see Figure 2)[2]. The staircase design will include a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

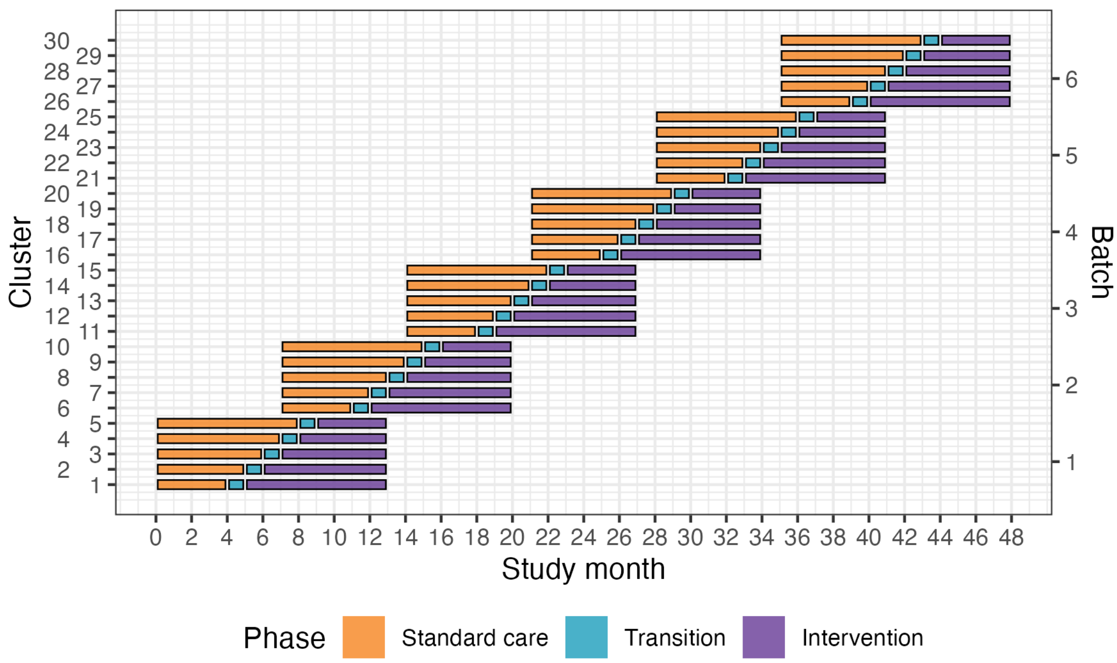

Figure 1: Trial design. Lines represent the duration of patient enrolment across clusters and phases. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.

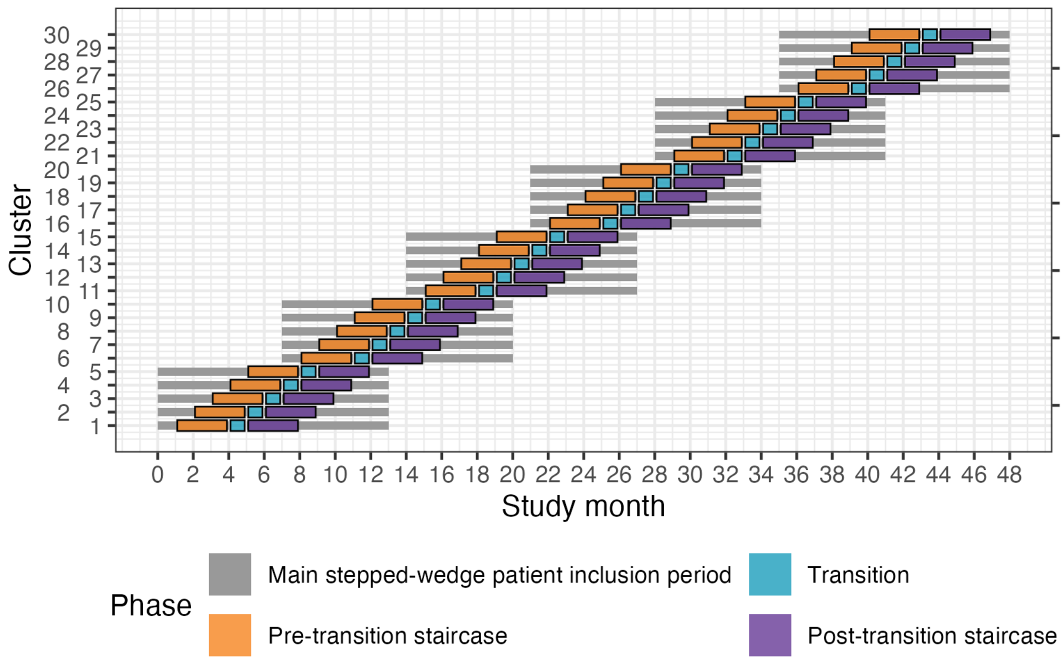

Figure 2: Nested staircase design. The design includes a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

5.2 Randomisation of clusters

We will enrol and assign clusters to batches once eligibility is confirmed and ethical approval is obtained. Each batch will include hospitals from different regions to optimise trial logistics and training delivery. Within each batch, clusters will be randomised to the intervention implementation sequences allocated to that batch.

Randomisation is conducted at the cluster level and stratified by batch. Within each batch, allocation is balanced on cluster size, defined as the monthly volume of eligible patients, using covariate-constrained randomisation (CCR). CCR was selected to improve balance between intervention conditions on this key design variable.

For example, in Batch 1, five clusters were randomised to one of five sequences using CCR. A statistician (JM) implemented a computer algorithm in R to evaluate all possible allocations (n = 120) and calculate a balance metric reflecting how evenly cluster size would be distributed between intervention and control conditions. For each allocation, the squared difference in anticipated cluster size between conditions was calculated; because only one covariate was used, this squared difference constituted the balance statistic.

The 10% of allocations demonstrating the greatest imbalance were excluded, and the remaining allocations formed the candidate set. The final allocation was then selected at random from this set. Owing to limitations in the algorithm, the balance score was calculated assuming a complete design, although the trial design includes transition periods. This CCR procedure is repeated independently for each batch.

The random allocation sequence is generated by JM and communicated to the trial team to allow logistical arrangements, including securing places on upcoming ATLS® training courses, to be initiated. Allocation order will be concealed from participating sites for as long as logistically feasible, recognising that advance planning is required to release physicians for training.

Random selection of patients for nested staircase outcome collection is separate from cluster randomisation and is described under Nested staircase design and secondary outcomes (stratified random sampling by shift).

5.3 Outcomes

A summary of all outcomes, analysis sets, effect measures, and handling of intercurrent events is provided in Table 1.

5.3.1 Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. We define this outcome as any death occurring in the hospital during the initial admission, up to 30 days post arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

5.3.2 Secondary outcomes

5.3.2.1 Collected using the main stepped-wedge design

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department. In contrast to the primary outcome, all-cause mortality includes all deaths regardless of setting. Data on this outcome will be collected in the same way as for the primary outcome.
- Length of emergency department stay. Data on this outcome will be collected from patient hospital records.
- Length of hospital stay. Data on this outcome will be collected from patient hospital records.
- Intensive care unit admission. Data on this outcome will be collected from patient hospital records.
- Length of intensive care unit stay. Data on this outcome will be collected from patient hospital records.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

5.3.2.2 Collected using the nested staircase design

- Adherence to ATLS® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS® primary survey, which was modelled based on previous work on ATLS® adherence[3]. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients. The sampling will be designed as a nested staircase design.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D5L. This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. We will collect this data using a nested staircase design.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. This data will also be collected using a nested staircase design.

Table 1: Summary of outcomes, analysis sets, effect measures, and handling of intercurrent events

Outcome	Design component	Data type	Analysis set	Main analysis model	Effect measure	Potential intercurrent events	Strategy for handling intercurrent events
Primary outcome: In-hospital mortality within 30 days of arrival at the emergency department	Main stepped-wedge	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD	Loss to follow-up; withdrawal	Deaths after transfer included in outcome
All-cause mortality within 24 hours of arrival at the emergency department	Main stepped-wedge	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD	Loss to follow-up; withdrawal	All deaths regardless of setting
All-cause mortality within 30 days of arrival at the emergency department	Main stepped-wedge	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD	Loss to follow-up; withdrawal	All deaths regardless of setting
All-cause mortality within three months of arrival at the emergency department	Main stepped-wedge	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	OR; ARD	Loss to follow-up; withdrawal	All deaths regardless of setting
Length of emergency department stay	Main stepped-wedge	Count (hours)	Primary analysis set	Mixed-effects negative binomial (log)	Rate ratio	Death; transfer	Observed hours to ED exit; death/transfer truncate stay
Length of hospital stay	Main stepped-wedge	Count (days)	Primary analysis set	Mixed-effects negative binomial (log)	Rate ratio	Death; transfer	Observed days; death/transfer truncate stay
Intensive care unit (ICU) admission	Main stepped-wedge	Dichotomous	Primary analysis set	Mixed-effects binomial (logit)	OR	Death; transfer	Classified by ICU admission before death/transfer
Length of intensive care unit stay	Main stepped-wedge	Count (days)	ICU admissions, primary analysis set	Mixed-effects negative binomial (log)	Rate ratio	Death; transfer	Observed days among ICU admissions; death/transfer truncate stay
Return to work at 30 days after arrival at the emergency department	Main stepped-wedge	Dichotomous	Alive at 30 days, outcome relevant	Mixed-effects binomial (logit)	OR	Death; loss to follow-up; not relevant	Survivors only
Return to work at three months after arrival at the emergency department	Main stepped-wedge	Dichotomous	Alive at 3 months, outcome relevant	Mixed-effects binomial (logit)	OR	Death; loss to follow-up; not relevant	Survivors only
Adherence to ATLS® principles during initial patient resuscitation	Nested staircase	Proportion	Observed nested sample	Mixed-effects beta (logit)	Logit-scale difference	Not observed; incomplete window	Observed sample
Quality of life within seven days of discharge	Nested staircase	Ordinal EQ-5D-5L; VAS	Alive, nested sample	Cumulative logit; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only
Quality of life at 30 days after arrival at the emergency department	Nested staircase	Ordinal EQ-5D-5L; VAS	Alive at 30 days, nested sample	Cumulative logit; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only
Quality of life at three months after arrival at the emergency department	Nested staircase	Ordinal EQ-5D-5L; VAS	Alive at 3 months, nested sample	Cumulative logit; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only
Disability within seven days of discharge	Nested staircase	Ordinal WHO-DAS 2.0; summary score	Alive, nested sample	Ordinal mixed; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only
Disability at 30 days after arrival at the emergency department	Nested staircase	Ordinal WHO-DAS 2.0; summary score	Alive at 30 days, nested sample	Ordinal mixed; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only

(continued)

Outcome	Design component	Data type	Analysis set	Main analysis model	Effect measure	Potential intercurrent events	Strategy for handling intercurrent events
Disability at three months after arrival at the emergency department	Nested staircase	Ordinal WHODAS 2.0; summary score	Alive at 3 months, nested sample	Ordinal mixed; linear mixed	COR; mean difference	Death; loss to follow-up	Survivors only

5.4 Statistical hypotheses

The primary analysis will estimate the effect of ATLS training on 30-day in-hospital mortality compared with standard care. The treatment effect will be expressed as an odds ratio (OR) and as an absolute risk difference (ARD). An OR < 1 or ARD < 0 indicates lower mortality among patients treated in hospitals after ATLS implementation compared with standard care. Based on our pilot study and systematic review of the literature, we expect the intervention to reduce mortality.

5.5 Sample size calculations

Sample size calculations were performed during the trial design phase using the Shiny CRT Calculator[4].

5.5.1 Main stepped wedge design and primary outcome

With 30 clusters across 6 batches and a total sample size of 4320 our study has ~90% power across different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC) to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training (see Figure 3 A)[4]. This 5% absolute reduction in mortality is a clinically meaningful and conservative estimate based on our updated systematic review and pilot study[5,6]. We allowed for the clustered design, incorporated the one month transition period, and assumed: a discrete time decay correlation structure, an ICC of 0.02 (but considered sensitivity across the range 0.01-0.05), and a CAC of 0.9 (but considered sensitivity across the range 0.8-1.0), based on our pilot study and current guidance[7–10]. We included the CAC to allow for variation in clustering over time. We assume that each cluster will contribute approximately 12 observations per month to the analysis, but allowed for substantial variation in cluster sizes, based on our previous work. We also estimated the sample size required to detect a reduction in the primary outcome from 10% to 7.5%, which would require at least 30 observations per period and cluster (see Figure 3 B).

5.5.2 Nested staircase design and secondary outcomes

Adherence, quality of life, and disability collected in the nested staircase are secondary outcomes; the nested power calculations below inform only the sample size required for their data collection.

The secondary outcomes that will be measured using the nested staircase design are secondary outcomes, and the power calculations presented here are intended only to inform the sample size required for their data collection within the nested staircase design. These outcomes are adherence to ATLS® principles during initial patient resuscitation, quality of life, and disability. The expected effect of the intervention on each of these outcomes are an improvement in adherence from 50% during standard care to 70% after training[11], an increase in EQ5D5L health status from 70 during care to 75 after training[12], and finally a decrease in disability from a baseline value of 25 during standard care to 22.5 after training[13]. For quality of life and disability, these effects correspond to standardised effect sizes, expressed as Cohen's d, of 0.5. With 30 clusters in total, six per sequence, for a discrete time decay correlation structure, with ICCs of 0.01 to 0.15 and a CAC of 0.8, there is >80% power to detect these effects by including four patients in each cluster in each period. Accounting for loss to follow-up, we will include at least six patients per cluster per month. These patients will be a random subset of patients included in the

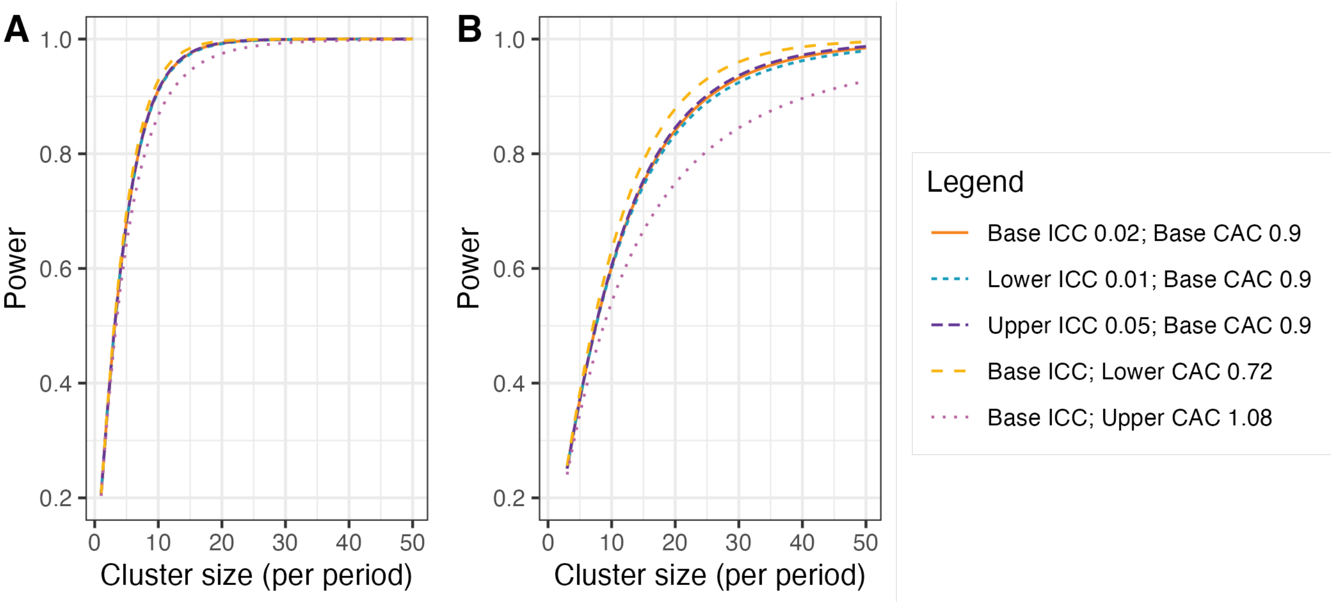

Figure 3: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC). **A)** Shows power curves assuming a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training. **B)** Shows power curves assuming a reduction in the primary outcome from 10% under standard care to 7.5% after ATLS® training. Under this scenario, we would need to increase the sample size per month to around 30 observations to achieve 90% powere under most combinations of CAC and ICC.

primary outcome data collection during the staircase months. The random subset will be selected using stratified random sampling by shift, meaning that the timing of the clinical research coordinator's shift will be randomised to cover approximately eight hours during the morning, afternoon, or night shift. Adherence to ATLS® principles, quality of life, and disability will be measured in all patients included during these shifts. In each hospital, the number of shifts that will be randomised will be determined by the volume of patients included during the months preceding the staircase months.

5.6 Statistical principles

5.6.1 Statistical software

We will use the R Statistical Software for all analyses[14].

5.6.2 Levels of statistical significance and confidence

We will use a two-sided significance level of 0.05 for all analyses, and we will report 95% confidence intervals (CI) for all estimates. We will not adjust for multiple testing because no secondary outcome is regarded as singularly more important.

Primary and secondary outcomes will be analysed collectively in a single final analysis once all trial data are available.

5.7 Analysis sets

The trial has a hierarchical structure in which hospitals represent the clusters, units within hospitals deliver care, and individual patients constitute the unit of observation and analysis. Randomisation is performed at the hospital level, with all trained units within the same hospital treated as part of the same cluster. Analysis models therefore account for clustering at the hospital level and do not include a separate random effect for unit within hospital. The primary analysis set will include all observations within clusters with available data on the primary outcome and irrespective of the receipt of the intervention.

Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. All data will be included with the exception of the transition phases. Group allocation for a patient depends on the period in which the patient was admitted to the hospital. If clusters deviate considerably from the scheduled date of transition, defined as more than four weeks, we will conduct a sensitivity analysis using the actual date of transition to define exposure.

The analysis set will be specified for each analysis. For the primary outcome, mortality secondary outcomes, length-of-stay outcomes, and ICU admission, the analysis set is all eligible patient participants included in the main stepped-wedge trial (with length of ICU stay restricted to patients admitted to ICU). For quality-of-life and disability outcomes, the analysis set is patients alive at the relevant follow-up time point within the nested staircase sample. For return-to-work outcomes, the analysis set is patients alive at the relevant follow-up time point in the main stepped-wedge trial for whom return to work is relevant. For adherence to ATLS® principles, the analysis set is the random sample of patients observed within the nested staircase design.

The analysis set for each outcome will include all available participants from the relevant target population with observed outcome data, subject to any restrictions imposed by the outcome definition or the nested staircase sampling scheme. For adjusted analyses under the available-case approach, the analysis set will be restricted to participants with complete covariate data, as described in Section 5.10.4.

We will use a CONSORT diagram to display the flow of hospitals, clusters and patients through the trial, see Figure 4. We will present cluster level summaries of the intervention effect. We will report the study according to the CONSORT guidelines for stepped-wedge randomised trials[15].

5.8 Baseline analyses

5.8.1 Cluster characteristics

We will describe cluster characteristics including location and size using frequencies and percentages for discrete variables and medians and interquartile ranges (Q1-Q3) for continuous variables. See Table 2 for details.

Table 2: Cluster characteristics

	Implementation sequence					
Characteristic	**Sequence 1**	**Sequence 2**	**Sequence 3**	**Sequence 4**	**Sequence 5**	**Overall**
Monthly trauma patient volume, Median (Q1, Q3)						
Hospital beds, Median (Q1, Q3)						
Intensive care unit beds, Median (Q1, Q3)						
Dedicated trauma beds, Median (Q1, Q3)						
Specialities available around the clock, n (%)						
General surgery						
Orthopaedics						
Neurosurgery						
Vascular surgery						
Interventional radiology						
Emergency medicine						
Facilities available around the clock, n (%)						
X-ray						
Ultrasound/FAST						
CT						
MRI						
Blood bank						
Initial resuscitation provider, n (%)						
Casualty medical officers						
Emergency medicine residents						
Surgical residents						

Figure 4: Consort diagram. The diagram displays the flow of clusters and participants through the trial.

5.8.2 Patient characteristics

We will describe patient characteristics among all eligible patients over the entire trial period, stratified by before and after ATLS training and overall, using frequencies and percentages for discrete variables and medians and interquartile ranges (Q1-Q3) for continuous variables. Missing values will be reported for each characteristic. We will not adjust for clustering when presenting baseline characteristics (i.e. non-outcome variables). We will present key characteristics in the results section, and present all characteristics as supplementary tables. See Table 3 for key characteristics and Table 7 for all characteristics.

Table 3: Key patient characteristics

	ATLS training		
Characteristic	**Before ATLS**	**After ATLS**	**Overall**
Age (years), Median (Q1, Q3)			
Missing			
Sex, n (%)			
Female			
Male			
Other			
Missing			
Transferred in, n (%)			
Missing			
Mechanism of injury, n (%)			
Road traffic injury			
Fall			
Assault			
Other			
Missing			
Injury Severity Score, Median (Q1, Q3)			
Missing			
Glasgow Coma Scale score, Median (Q1, Q3)			
Missing			
Systolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Respiratory rate (breaths/min), Median (Q1, Q3)			
Missing			
Oxygen saturation (%), Median (Q1, Q3)			
Missing			
Surgery, n (%)			
Missing			
Transfusion, n (%)			
Missing			
Imaging, n (%)			
Ultrasound			
X-ray			
CT			
Missing			

5.8.3 Outcomes

Summaries of primary and secondary outcomes will be presented in the results section, stratified by before and after ATLS training and overall, using the same summary statistics as the patient characteristics tables. Missing values will be reported for each outcome. We will present key outcome summaries in the results section, and present all outcome summaries as supplementary tables. See Table 4 for key outcomes and Table 8 for all outcomes.

Table 4: Descriptive summaries of key outcomes

	ATLS training		
Outcome	**Before ATLS**	**After ATLS**	**Overall**
Primary outcome			
In-hospital mortality within 30 days, n (%)			
Missing			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, n (%)			
Missing			
All-cause mortality within 30 days, n (%)			
Missing			
All-cause mortality within three months, n (%)			
Missing			
Length of emergency department stay (hours), Median (Q1, Q3)			
Missing			
Length of hospital stay (days), Median (Q1, Q3)			
Missing			
Intensive care unit admission, n (%)			
Missing			
Length of intensive care unit stay (days), Median (Q1, Q3)			
Missing			
Return to work at 30 days, n (%)			
Missing			
Return to work at three months, n (%)			
Missing			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS principles (%), Median (Q1, Q3)			
Missing			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			

5.9 Interim analysis

There will be one interim analysis after half of the batches have completed the trial. The interim analysis will be assessed by the joint Trial Steering and Data Monitoring Committee. The purposes of this interim analysis will be to assess the trial's feasibility and recommend that the trial be stopped if it is not feasible (e.g. if hospitals fail to adhere to the randomisation schedule or if there are substantial missing data in outcomes), and to compare characteristics across intervention conditions to monitor for differential recruitment/ascertainment between the intervention and control groups. No efficacy stopping is planned because any estimates obtained at the time point of the interim analysis will be associated with major statistical uncertainties.

The Trial Team will prepare the interim analysis report for the joint Trial Steering and Data Monitoring Committee (SDMC), for approval by the Trial Management Group before circulation. The report will follow the standard SDMC report template in the SDMC charter and will include:

- trial progress: numbers of clusters included and dropping out; numbers of potentially eligible participants screened, participants included, participants who did not consent to out-of-hospital follow-up, and participants lost to follow-up, overall and by intervention condition;
- adherence to the randomisation schedule and reasons for non-adherence;
- summary statistics of included clusters and patient participants by intervention condition, to monitor for differential recruitment or ascertainment;
- completeness of primary and secondary outcome data, overall and by intervention condition, without presenting outcome values by intervention condition;
- ATLS® training delivery, including numbers of physicians trained in each hospital;
- protocol deviations and proposed protocol amendments;
- safety events, as defined for SDMC monitoring;
- relevant external evidence; and
- the SDMC recommendation.

No formal interim efficacy analysis or efficacy-based stopping rule will be applied. Outcomes will not be compared across intervention conditions in the interim report, to avoid biasing the SDMC or Trial Management Group. The SDMC will use the interim report to advise whether the trial remains feasible and should continue as planned.

5.10 Analysis of the primary outcome

The primary outcome is in-hospital mortality within 30 days of arrival at the emergency department and will be analysed as a dichotomous variable. We will estimate the intervention effect as the odds ratio (OR) of death between the ATLS® and standard care arms, with an OR < 1 indicating lower odds of death in the ATLS® arm compared to the standard care arm and vice versa. We will also estimate the absolute risk difference (ARD) of death between the ATLS® and standard care arms, with an ARD < 0 indicating lower risk of death in the ATLS® arm compared to the standard care arm and vice versa.

We will use a mixed-effects binomial regression model to estimate the effect of ATLS® on mortality. The OR will be estimated using the logit link, and the ARD will be estimated using the identity link. These models account for clustering at the hospital level and provide cluster-specific estimates of the intervention effect. The primary analysis will be based on the most complex model that converges from a prespecified sequence of models of decreasing complexity. The analysis will proceed from the most complex model to progressively simpler models if there are convergence or numerical stability issues. The primary analysis will then be subject to sensitivity analyses. If the binomial model with the logit link converges but the corresponding model with the identity link does not, then we will report only an OR.

All models in this sequence will include a fixed effect for intervention exposure and adjust for time, although the parametrisation of time and the random effects structure may be simplified.

The model sequence is defined as follows:

1. **Primary model**: a model including fixed effects for period, modelled as a categorical variable with separate period effects for each batch, and a fixed effect for intervention exposure. Clustering will be accounted for by including a random intercept for cluster and a random cluster-by-period effect, both assumed to follow a normal distribution. The model is specified in Equation 1.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_b t + \theta X_b kt + \alpha_b k + \gamma_b kt \tag{1}$$

2. **Simplified random effects structure**: a model including fixed effects for period and intervention exposure and a random intercept for cluster only, thereby removing the random cluster-by-period effect. The model is specified in Equation 2.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_b t + \theta X_b kt + \alpha_b k \tag{2}$$

3. **Model with shared period effects**: a model in which fixed period effects are shared across batches, with time adjusted for using a common period effect or calendar time modelled directly (e.g., as a continuous covariate or using splines). A random intercept for cluster and a random cluster-by-period effect are retained; clusters remain indexed by batch, but the fixed period effects are common across batches. The model is specified in Equation 3.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{3}$$

4. **Combined simplified model**: a model that includes both a simplified random effects structure (random intercept for cluster only) and shared period effects. This represents the most parsimonious mixed-effects model considered acceptable for the primary analysis. The model is specified in Equation 4.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} \tag{4}$$

Where:

- $g(\cdot)$: the link function, specified as either the logit link, $g(p) = \log\{p/(1-p)\}$, or the identity link, $g(p) = p$, depending on the estimand of interest.
- θ: **the intervention effect**, represented as a fixed effect of intervention exposure (the primary estimand); under the logit link this represents the log-odds ratio for death associated with ATLS® exposure compared with standard care, and under the identity link it represents the absolute risk difference.
- $\Pr(Y_{bkti} = 1)$: the probability of death for patient $i = 1, \dots, m$ in cluster $k = 1, \dots, 30$ during period $t = 1, \dots, 12$ in batch $b = 1, \dots, 6$.
- μ: the intercept, representing the value of the linear predictor for the reference levels of all fixed effects, with random effects equal to zero.
- β_t: the fixed effect of period t, shared across batches.
- β_{bt}: the fixed effect of period t in batch b, accounting for a separate period effect for each batch, resulting in a total of $6 \times (12)$ period effects.
- X_{bkt}: the intervention exposure indicator for cluster k during period t in batch b, with $X_{bkt} = 1$ for ATLS® and $X_{bkt} = 0$ for standard care.
- α_{bk}: a normally distributed random effect for cluster k in batch b, representing between-cluster (hospital-level) variability in the outcome.
- γ_{bkt}: a normally distributed random effect for cluster k in period t in batch b, representing within-cluster variation over time and capturing cluster-specific deviations from the average period effect.

The distributional assumptions for the random effects are specified in Equation 5.

$$\begin{aligned} \alpha_{bk} &\overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\alpha^2), \\ \gamma_{bkt} &\overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\gamma^2), \\ \alpha_{bk} &\perp \gamma_{bkt}. \end{aligned} \tag{5}$$

To correct potential inflation of the type I error rate due to the small number of clusters, a small-sample correction will be applied. Our primary method will be the Kenward–Roger correction[16], but this choice may be updated based on methodological developments available closer to the end of patient inclusion. The correction applied may also differ for outcomes collected via the complete stepped-wedge design and the incomplete nested staircase designs.

Standard errors will be model-based. Small-sample adjustments, including degrees-of-freedom corrections such as Kenward–Roger, will be applied where relevant. Confidence intervals will be Wald-type.

The general estimation framework for the mixed-effects models is maximum likelihood, with a Laplace approximation used to integrate out the random effects. The specific R package and numerical implementation will be chosen based on convergence behaviour and the availability of appropriate small-sample corrections closer to the time of analysis. If the binomial model with the logit link converges but the corresponding model with the identity link does not, then we will report only an OR.

Model diagnostics will be used to assess distributional assumptions and model convergence.

We will report time-adjusted within-cluster correlations for all outcomes with 95% CIs, including the intra-cluster correlation (ICC) and within- and between-period correlations implied by the assumed correlation structures, together with all estimated variance components. For binary outcomes, these correlations will additionally be reported on the latent scale.

Intra-cluster and related correlation parameters will be derived from the estimated variance components of the fitted models. For models with an AR(1) within-cluster structure, the correlation parameter will be estimated directly from the AR(1) structure.

5.10.1 Sensitivity analyses

We will conduct sensitivity analyses to assess the robustness of the primary analysis results to different modelling assumptions. In all sensitivity analyses, the focus will be on the stability of the estimated intervention effect θ. Unless otherwise stated, each sensitivity analysis will be operationalised using two separate models, one with the logit link and one with the identity link.

5.10.1.1 Models with autoregressive within-cluster correlation

We will explore whether models with more complex correlation structures provide a better fit to the data. These models are not used as primary analysis models because there is limited understanding of when such models converge and how best to choose between different plausible correlation structures. Instead, we include these models as sensitivity analyses.

We will first explore whether allowing temporal correlation of outcomes within clusters affects inference on the intervention effect. Specifically, we will consider a discrete time-decay correlation structure by specifying a random cluster effect with an autoregressive structure of order one (AR(1)), described in Equation 6. This parameterisation assumes that correlations between cluster-level random effects decay geometrically with increasing temporal separation between periods.

$$\alpha_{bk,t} = \rho\alpha_{bk,t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\alpha^2) \tag{6}$$

Where:

- ρ is the correlation parameter governing the association between adjacent periods within the same cluster.
- $\alpha_{bk,t}$ is the random effect for cluster k in period t in batch b.
- ϵ_t is a normally distributed error term.

We note that this AR(1) formulation differs slightly from alternative parameterisations used elsewhere in which correlations are specified through a joint covariance matrix. Although both approaches capture decaying temporal correlation, the parameter ρ is not directly equivalent to parameters used in other specifications and should not be interpreted interchangeably.

5.10.1.2 Models with random cluster by intervention effects

To assess sensitivity to the assumption of a common intervention effect across clusters, we will extend the primary analysis model to allow the intervention effect to vary by cluster. Specifically, we will include a random cluster-level slope for the intervention exposure, with a non-zero covariance between this slope and the cluster random intercept (as anticipated in the protocol). The cluster-by-period random effect remains independent of both. The model is specified in Equation 7.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + u_{bk} X_{bkt} \tag{7}$$

Where:

- u_{bk} is a random effect representing cluster-specific deviations from the average intervention effect.
- The cluster intercept and intervention slope are jointly normally distributed,

$$\begin{pmatrix} \alpha_{bk} \\ u_{bk} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \sigma_{\alpha u} \\ \sigma_{\alpha u} & \sigma_u^2 \end{pmatrix} \right),$$

  with $\sigma_{\alpha u} \neq 0$ allowed.
- $\gamma_{bkt} \sim \mathcal{N}(0, \sigma_\gamma^2)$ is independent of (α_{bk}, u_{bk}).

5.10.1.3 Models with time modelled with a spline function

We will further explore the potential for a time-varying treatment effect[17]. To explore if the fixed period effect is both parsimonious and adequate to represent the extent of any underlying secular trend, we will model calendar time using a shared natural cubic spline basis across all batches with knots at the equally spaced time points 3, 6 and 9. This will result in five spline basis functions, because the natural cubic splines are modelled with three degrees of freedom but are constrained to be linear before the first and after the last knot. The model is specified in Equation 8.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \sum_{j=1}^{5} \beta_j S_j(t, \{3, 6, 9\}) + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{8}$$

Where:

- t denotes calendar time (period index) common to all batches.
- $S_j(t, \{3, 6, 9\})$ denotes the jth natural cubic spline basis function, shared across batches.
- β_j is the coefficient for the jth spline basis function.

5.10.1.4 Models exploring lag and weaning effects

We will also extend the primary analysis model to include an interaction between treatment and number of periods since first treated, to examine if there is any indication of a relationship between duration of exposure to the intervention and outcomes. This will allow us to model different lag effects (whereby it takes time for the intervention to become embedded within the culture before its impact can properly start to be realised); as well as weaning effects (whereby the effect of the intervention starts to decrease – or fade). This type of analysis attempts to disentangle how some clusters end up having a long exposure to the intervention and others have a much shorter exposure time. The model is specified in Equation 9.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \theta_{\text{int}} X_{bkt} \times T_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{9}$$

Where θ_{int} represents the interaction between intervention exposure and time since first treated, and T_{bkt} is the number of periods since first exposure.

5.10.2 Adjusted analyses

Fully adjusted covariate analysis will additionally adjust for (expected % missing data in parentheses):

- Age (<5%)
- Sex (<5%)
- Systolic blood pressure (25%)
- Glasgow Coma Scale (25%)
- Injury Severity Score (10%)
- Mechanism of injury (10%)

These are known individual-level prognostic factors for the primary outcome. These covariates will be included in the models specified in Equation 1 as fixed effects. We will model the continuous covariates assuming linear effects. We will assume no interaction between subgroups and secular time trends, which may be evaluated in future analyses if indicated.

5.10.3 Subgroup analyses

We will perform the following subgroup analyses of the primary outcome:

- geographical region, defined using the state in which the participating hospital is located, with each state as a separate category; the number of states depends on the final set of participating clusters and cannot be prespecified;

- age groups, defined as older adolescents (15-19 years), young adults (20-24 years), adults (25-59 years), and older adults (60 years and older)[18];
- sex, using the levels male and female;
- clinical cohorts, defined as blunt multisystem trauma, penetrating trauma, and severe isolated traumatic brain injury, with modification to avoid overlap between the cohorts[19]; this subgroup analysis will include only patients classified into one of these three cohorts;
- patients with major trauma, defined as patients with an injury severity score (ISS) of 16 or higher versus those with ISS below 16;
- cluster size, defined as the monthly volume of eligible patients (as used for covariate-constrained randomisation) and categorised as small (fewer than 12 patients per month), medium (12–20), or large (more than 20)

These subgroup analyses will be conducted by adding the subgroup variable and the interaction between the subgroup variable and the intervention exposure variable as fixed effects to the models specified in Equation 1.

5.10.4 Treatment of missing data

We will present the frequency and percentage of missing data for all variables. Handling of missing data for the primary analysis will depend on the percentage of missing data for the primary outcome.

If less than 10% of participants have a missing primary outcome, we will use an available-case approach. The primary unadjusted analysis will include all participants with observed primary outcome data. Adjusted analyses will be restricted to participants with complete data on the adjustment covariates, and the unadjusted analysis will be repeated in this complete-case population to allow the results of the unadjusted and ajusted analyses to be directly compared.

If 10% or more of participants have a missing primary outcome, we will use multiple imputation by chained equations (MICE) under a missing-at-random assumption, imputing the primary outcome, secondary outcomes as required for analysis, and the covariates included in the fully adjusted model, with a minimum of 20 imputations. The imputation model will reflect the hierarchical data structure, including clustering at the hospital level and design variables such as period and intervention exposure. Unadjusted and adjusted analyses will then be conducted using the imputed datasets. We will also present the corresponding complete-case analyses for comparison. Secondary outcomes will not be imputed under the available-case approach.

If there is evidence that data are missing not at random, we will explore this in a sensitivity analysis.

5.11 Analysis of secondary outcomes

Unless otherwise stated, secondary outcomes will be analysed using the same mixed-effects modelling framework as specified for the primary outcome, including fixed effects for time and intervention exposure and random effects for cluster and cluster-by-period, with the link function and outcome distribution adapted to the scale of the outcome. Effect measures are odds ratios and absolute risk differences for binary outcomes, rate ratios for count (length-of-stay) outcomes, cumulative odds ratios for ordinal outcomes, mean differences for continuous outcomes, and a logit-scale difference for the adherence proportion; there are no time-to-event analyses and no hazard ratios. These models provide cluster-specific estimates of the intervention effect for patient-level outcomes. Model diagnostics will be used to assess distributional assumptions, the proportional-odds assumption for cumulative logit models, and model convergence.

5.11.1 All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department

These outcomes will be analysed as dichotomous variables using the mixed-effects binomial model with the logit and identity links specified in Equation 1, to estimate odds ratios and absolute risk differences.

5.11.2 Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

Quality of life will be measured within the nested staircase design using the official and validated translations of the EQ-5D-5L. The EQ-5D-5L yields two outcome types: five ordinal domain scores (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), each rated on a Likert scale from 1 to 5, and a visual analogue scale (VAS) ranging from 0 to 100. For these analyses, period effects are defined on the global study timeline (not separately within each batch), and not all clusters contribute data in every period.

For each of the five ordinal domains, we will use a mixed-effects cumulative logit (proportional odds) model:

$$\text{logit}\{\Pr(Y_{bkti} \leq j)\} = \mu_j + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{10}$$

Where μ_j is the intercept for the jth category.

The VAS will be analysed as a continuous outcome using a linear mixed-effects model:

$$\text{VAS}_{bkti} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + \epsilon_{bkti}, \quad \epsilon_{bkti} \sim \mathcal{N}(0, \sigma^2) \tag{11}$$

5.11.3 Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

Disability will be measured within the nested staircase design using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0)[20]. This tool assesses six domains of functioning: cognition, mobility, self-care, getting along, life activities, and participation. Each domain is rated on a likert scale from 1 to 5, with 1 indicating no difficulties and 5 indicating extreme difficulties. As for quality of life, period effects are defined on the global study timeline, and not all clusters contribute data in every period. We will analyse each domain separately using a mixed effects ordinal model as specified in Equation 10. We will also calculate a WHODAS 2.0 summary score using the method referred to as the "complex scoring" method. This method involves summing the item scores within each of the six domains, then summing the scores of all domains, and finally transforming the total score to a 0-100 scale. We will analyse the summary score using a linear mixed effects model as specified in Equation 11.

5.11.4 Return to work at 30 days and three months after arrival at the emergency department, among survivors

Return to work will be analysed as a dichotomous outcome using the mixed-effects binomial model specified in Equation 1 with the logit link.

5.11.5 Length of emergency department stay

Length of emergency department stay will be analysed as a count outcome: the number of hours from arrival at the emergency department until ED exit. Exit may occur via ward admission, ICU admission, transfer to another hospital, death, or discharge home; all of these routes contribute the observed ED duration. The purpose of the analysis is to estimate the effect of ATLS® on mean ED length of stay. Death and transfer truncate the observed stay and are not treated as censoring; mortality is analysed separately as a primary and secondary outcome.

We will use a mixed-effects negative binomial regression model with a log link, including fixed effects for intervention exposure and batch-specific categorical period effects, a random intercept for cluster, and a random cluster-by-period effect, matching the primary analysis model structure where feasible. If this model does not converge, we will simplify the random-effects and period structure following the same sequence as for the primary outcome. The model is specified in Equation 12. The intervention effect θ is the log rate ratio for ED length of stay associated with ATLS® exposure compared with standard care; we will also report the corresponding multiplicative effect on the mean count (rate ratio).

$$\log\{\mathbb{E}(Y_{bkti})\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{12}$$

Where Y_{bkti} is the length-of-stay count for patient i in cluster k during period t in batch b, and all other terms are defined as in the primary analysis model. A dispersion parameter will be estimated as part of the negative binomial model.

5.11.6 Length of hospital stay

Length of hospital stay will be analysed as a count outcome: the number of days from arrival at the emergency department to hospital discharge (or death in hospital). The analysis uses the mixed-effects negative binomial model in Equation 12, with the same handling of death and transfer as for ED length of stay.

5.11.7 Intensive care unit admission

Intensive care unit admission will be analysed as a dichotomous outcome using the mixed-effects binomial model specified in Equation 1 with the logit link. Patients who die or are transferred without ICU admission will be classified according to whether ICU admission occurred before that event.

5.11.8 Length of intensive care unit stay

Length of intensive care unit stay will be analysed as a count outcome among patients admitted to ICU: the number of days from ICU admission to ICU discharge (or death in ICU). The analysis uses the mixed-effects negative binomial model in Equation 12, with the same handling of death and transfer as for the other length-of-stay outcomes.

5.11.9 Adherence to ATLS® principles during initial patient resuscitation

We will assess adherence to ATLS® principles within the nested staircase design, in both the standard care and ATLS® periods, using a 14-item checklist that covers the key steps of the ATLS® primary survey. Adherence will be measured as the percentage of completed checklist items (completion of all 14 steps corresponds to 100% adherence). For these analyses, period effects are defined on the global study timeline (not separately within each batch), and not all clusters contribute data in every period. This outcome will be analysed using a mixed-effects beta regression model with a logit link, with the percentage first expressed as a proportion on the 0–1 scale:

$$g\{\mathbb{E}(Y_{bkti})\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \tag{13}$$

Where $\mathbb{E}(Y_{bkti})$ denotes the expected adherence proportion. The intervention effect θ represents the difference in expected adherence proportion on the logit (link) scale. Because the observed adherence proportion may take values of 0 or 1, values will be rescaled to lie in the open interval $(0, 1)$ prior to fitting the beta regression model. A beta regression precision (dispersion) parameter will be estimated as part of the model.

5.12 Presentation of analysis results

Intervention-effect estimates for the primary and secondary outcomes will be reported with 95% CIs and p-values, using the effect measures specified for each outcome. Results tables will be prepared in the style of `gtsummary::tbl_regression`, using the same section headers and label style as the descriptive outcome summaries. We will present key outcome analyses in the results section, and present all outcome analyses as supplementary tables. See Table 5 for key analyses and Table 9 for all analyses.

Table 5: Shell table of key primary and secondary outcome analysis results

Outcome	**Estimate**	**95% CI**	**p-value**
Primary outcome			
In-hospital mortality within 30 days, Odds ratio			
In-hospital mortality within 30 days, Absolute risk difference			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, Odds ratio			
All-cause mortality within 24 hours, Absolute risk difference			
All-cause mortality within 30 days, Odds ratio			
All-cause mortality within 30 days, Absolute risk difference			
All-cause mortality within three months, Odds ratio			

(continued)

Outcome	Estimate	95% CI	p-value
All-cause mortality within three months, Absolute risk difference			
Length of emergency department stay, Rate ratio			
Length of hospital stay, Rate ratio			
Intensive care unit (ICU) admission, Odds ratio			
Length of intensive care unit stay, Rate ratio			
Return to work at 30 days, Odds ratio			
Return to work at three months, Odds ratio			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS® principles, Logit-scale difference			
Secondary outcomes within seven days of discharge (nested staircase design)			
Quality of life, Cumulative odds ratio			
Quality of life, Mean difference			
Disability, Cumulative odds ratio			
Disability, Mean difference			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
Quality of life, Cumulative odds ratio			
Quality of life, Mean difference			
Disability, Cumulative odds ratio			
Disability, Mean difference			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
Quality of life, Cumulative odds ratio			
Quality of life, Mean difference			
Disability, Cumulative odds ratio			
Disability, Mean difference			

Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio, OR = Odds Ratio

Sensitivity, fully adjusted, and subgroup analyses of the primary outcome will be reported in the same style. See Table 6.

Table 6: Shell table of sensitivity, fully adjusted, and subgroup analysis results for the primary outcome

Analysis	Estimate	95% CI	p-value
Sensitivity analyses			
Autoregressive within-cluster correlation (AR(1)), Odds ratio			
Autoregressive within-cluster correlation (AR(1)), Absolute risk difference			
Random cluster-by-intervention effects, Odds ratio			
Random cluster-by-intervention effects, Absolute risk difference			
Time modelled with a spline function, Odds ratio			
Time modelled with a spline function, Absolute risk difference			
Lag and weaning effects (intervention), Odds ratio			
Lag and weaning effects (intervention), Absolute risk difference			
Lag and weaning effects (periods since first exposure), Odds ratio			
Lag and weaning effects (periods since first exposure), Absolute risk difference			
Actual date of transition for intervention exposure, Odds ratio			
Actual date of transition for intervention exposure, Absolute risk difference			
Fully adjusted analysis			
Fully adjusted analysis, Odds ratio			
Fully adjusted analysis, Absolute risk difference			
Subgroup analyses			

(continued)

Analysis	Estimate	95% CI	p-value
Geographical region (state-specific estimates; states depend on participating clusters), Odds ratio			
Geographical region (state-specific estimates; states depend on participating clusters), Absolute risk difference			
Age group: Older adolescents (15-19 years), Odds ratio			
Age group: Older adolescents (15-19 years), Absolute risk difference			
Age group: Young adults (20-24 years), Odds ratio			
Age group: Young adults (20-24 years), Absolute risk difference			
Age group: Adults (25-59 years), Odds ratio			
Age group: Adults (25-59 years), Absolute risk difference			
Age group: Older adults (60 years and older), Odds ratio			
Age group: Older adults (60 years and older), Absolute risk difference			
Sex: Male, Odds ratio			
Sex: Male, Absolute risk difference			
Sex: Female, Odds ratio			
Sex: Female, Absolute risk difference			
Clinical cohort: Blunt multisystem trauma, Odds ratio			
Clinical cohort: Blunt multisystem trauma, Absolute risk difference			
Clinical cohort: Penetrating trauma, Odds ratio			
Clinical cohort: Penetrating trauma, Absolute risk difference			
Clinical cohort: Severe isolated traumatic brain injury, Odds ratio			
Clinical cohort: Severe isolated traumatic brain injury, Absolute risk difference			
Major trauma: ISS >=16, Odds ratio			
Major trauma: ISS >=16, Absolute risk difference			
Major trauma: ISS <16, Odds ratio			
Major trauma: ISS <16, Absolute risk difference			
Cluster size: Small (<12 patients/month), Odds ratio			
Cluster size: Small (<12 patients/month), Absolute risk difference			
Cluster size: Medium (12-20 patients/month), Odds ratio			
Cluster size: Medium (12-20 patients/month), Absolute risk difference			
Cluster size: Large (>20 patients/month), Odds ratio			
Cluster size: Large (>20 patients/month), Absolute risk difference			

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Time-adjusted within-cluster correlations and variance components will be reported with 95% CIs for all outcomes, as specified above; for binary outcomes, latent-scale correlations will additionally be reported. For models with an AR(1) within-cluster structure, the within-period correlation and correlation decay parameter will also be reported. These parameters will be presented as a supplementary table. See Table 10.

6 Supplementary tables

Table 7: All patient characteristics

	ATLS training		
Characteristic	**Before ATLS**	**After ATLS**	**Overall**
Age (years), Median (Q1, Q3)			
Missing			
Sex, n (%)			
Female			
Male			
Other			
Missing			
Marital status, n (%)			
Never married			
Currently married			
Separated			
Divorced			
Widowed			
Cohabiting			
Missing			
Education level, n (%)			
Not attended school			
Primary school			
Secondary school			
Higher secondary school			
Graduate			
Post graduate and above			
Other			
Missing			
Main work status, n (%)			
Paid work, such as daily wage earner, teacher, factory worker and government employee			
Self-employed, such as own your business or farming			
Non-paid work, such as volunteer or charity			
Student			
Keeping house/homemaker			
Retired			
Unemployed (health reasons)			
Unemployed (other reasons)			
Other			
No income			
Missing			
Income level (INR per month), n (%)			
Below 10,000			
10,001-20,000			
20,001-30,000			
30,001-50,000			
50,001-80,000			
80,001-1,00,000			
Above 1,00,000			
No income			
Missing			
Comorbidities (Charlson Comorbidity Index), n (%)			
Myocardial infarction			
Congestive heart failure			
Peripheral vascular disease			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Cerebrovascular disease			
Dementia			
Chronic pulmonary disease			
Rheumatologic disease			
Peptic ulcer disease			
Liver disease			
Diabetes			
Hemiplegia or paraplegia			
Renal disease			
Malignancy			
Leukemia			
Lymphoma			
AIDS			
None			
Other			
Missing			
Severity of liver disease, n (%)			
None			
Mild			
Moderate or severe			
Missing			
Severity of diabetes, n (%)			
None			
Controlled			
Uncontrolled			
Missing			
Severity of malignancy, n (%)			
None			
Localized			
Metastatic tumor			
Missing			
Clinical Frailty Scale, n (%)			
1. Very Fit			
2. Fit			
3. Managing well			
4. Living with very mild frailty			
5. Living with mild frailty			
6. Living with moderate frailty			
7. Living with severe frailty			
8. Living with very severe frailty			
9. Terminally ill			
Missing			
Mode of transport, n (%)			
Ambulance			
Police			
Private vehicle			
Walking			
Others			
Missing			
Transferred in, n (%)			
Missing			
Mechanism of injury, n (%)			
Road traffic injury			
Fall			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Assault			
Other			
Missing			
Injury Severity Score, Median (Q1, Q3)			
Missing			
Injury source data, n (%)			
Medical record			
X-ray report			
CT-report			
Surgical notes			
Missing			
Glasgow Coma Scale score, Median (Q1, Q3)			
Missing			
Systolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Diastolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Heart rate (beats/min), Median (Q1, Q3)			
Missing			
Respiratory rate (breaths/min), Median (Q1, Q3)			
Missing			
Oxygen saturation (%), Median (Q1, Q3)			
Missing			
Body temperature (°F), Median (Q1, Q3)			
Missing			
Emergency department disposition, n (%)			
Admitted			
Referred or transferred for admission			
Dead			
Others			
Missing			
Type of admitting ward, n (%)			
General surgery			
Orthopaedics			
Neurosurgery			
Intensive care unit			
High dependency unit			
Medicine			
Trauma ward			
Missing			
Hospital disposition, n (%)			
Alive			
Dead			
Transferred for admission			
Missing			
Transferred to another hospital, n (%)			
Missing			
Surgery, n (%)			
Missing			
Preoperative ASA score, n (%)			
1. A normal healthy patient			

(continued)

	ATLS training		
Characteristic	**Before ATLS**	**After ATLS**	**Overall**
2. A patient with mild systemic disease			
3. A patient with severe systemic disease			
4. A patient with severe systemic disease that is a constant threat to life			
5. A moribund patient who is not expected to survive without the operation			
6. A declared brain-dead patient whose organs are being removed for donor purposes			
Missing			
Transfusion, n (%)			
Missing			
Type of blood product, n (%)			
Packed red blood cells			
Platelets			
Fresh frozen plasma			
Whole blood			
Other			
Missing			
Number of units transfused, Median (Q1, Q3)			
Missing			
Imaging, n (%)			
Ultrasound			
X-ray			
CT			
Missing			

Table 8: Descriptive summaries of all outcomes

	ATLS training		
Outcome	**Before ATLS**	**After ATLS**	**Overall**
Primary outcome			
In-hospital mortality within 30 days, n (%)			
Missing			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, n (%)			
Missing			
All-cause mortality within 30 days, n (%)			
Missing			
All-cause mortality within three months, n (%)			
Missing			
Length of emergency department stay (hours), Median (Q1, Q3)			
Missing			
Length of hospital stay (days), Median (Q1, Q3)			
Missing			
Intensive care unit admission, n (%)			
Missing			
Length of intensive care unit stay (days), Median (Q1, Q3)			
Missing			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Return to work at 30 days, n (%)			
Missing			
Return to work at three months, n (%)			
Missing			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS principles (%), Median (Q1, Q3)			
Missing			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			

(continued)

	ATLS training		
Outcome	**Before ATLS**	**After ATLS**	**Overall**
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			

(continued)

Outcome	ATLS training: Before ATLS	ATLS training: After ATLS	Overall
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L index score, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			

Table 9: Shell table of all primary and secondary outcome analysis results

Outcome	Estimate	95% CI	p-value
Primary outcome			
In-hospital mortality within 30 days, Odds ratio			
In-hospital mortality within 30 days, Absolute risk difference			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, Odds ratio			
All-cause mortality within 24 hours, Absolute risk difference			
All-cause mortality within 30 days, Odds ratio			
All-cause mortality within 30 days, Absolute risk difference			
All-cause mortality within three months, Odds ratio			
All-cause mortality within three months, Absolute risk difference			
Length of emergency department stay, Rate ratio			
Length of hospital stay, Rate ratio			
Intensive care unit (ICU) admission, Odds ratio			
Length of intensive care unit stay, Rate ratio			
Return to work at 30 days, Odds ratio			
Return to work at three months, Odds ratio			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS® principles, Logit-scale difference			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L mobility, Cumulative odds ratio			
EQ-5D-5L self-care, Cumulative odds ratio			
EQ-5D-5L usual activities, Cumulative odds ratio			
EQ-5D-5L pain/discomfort, Cumulative odds ratio			
EQ-5D-5L anxiety/depression, Cumulative odds ratio			
EQ-5D-5L VAS, Mean difference			
WHODAS 2.0 cognition, Cumulative odds ratio			
WHODAS 2.0 mobility, Cumulative odds ratio			
WHODAS 2.0 self-care, Cumulative odds ratio			

(continued)

Outcome	Estimate	95% CI	p-value
WHODAS 2.0 getting along, Cumulative odds ratio			
WHODAS 2.0 life activities, Cumulative odds ratio			
WHODAS 2.0 participation, Cumulative odds ratio			
WHODAS 2.0 summary score, Mean difference			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L mobility, Cumulative odds ratio			
EQ-5D-5L self-care, Cumulative odds ratio			
EQ-5D-5L usual activities, Cumulative odds ratio			
EQ-5D-5L pain/discomfort, Cumulative odds ratio			
EQ-5D-5L anxiety/depression, Cumulative odds ratio			
EQ-5D-5L VAS, Mean difference			
WHODAS 2.0 cognition, Cumulative odds ratio			
WHODAS 2.0 mobility, Cumulative odds ratio			
WHODAS 2.0 self-care, Cumulative odds ratio			
WHODAS 2.0 getting along, Cumulative odds ratio			
WHODAS 2.0 life activities, Cumulative odds ratio			
WHODAS 2.0 participation, Cumulative odds ratio			
WHODAS 2.0 summary score, Mean difference			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L mobility, Cumulative odds ratio			
EQ-5D-5L self-care, Cumulative odds ratio			
EQ-5D-5L usual activities, Cumulative odds ratio			
EQ-5D-5L pain/discomfort, Cumulative odds ratio			
EQ-5D-5L anxiety/depression, Cumulative odds ratio			
EQ-5D-5L VAS, Mean difference			
WHODAS 2.0 cognition, Cumulative odds ratio			
WHODAS 2.0 mobility, Cumulative odds ratio			
WHODAS 2.0 self-care, Cumulative odds ratio			
WHODAS 2.0 getting along, Cumulative odds ratio			
WHODAS 2.0 life activities, Cumulative odds ratio			
WHODAS 2.0 participation, Cumulative odds ratio			
WHODAS 2.0 summary score, Mean difference			

Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio, OR = Odds Ratio

Table 10: Shell table of within-cluster correlation parameters for all outcomes

Parameter	Estimate	95% CI
In-hospital mortality within 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
All-cause mortality within 24 hours		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		

(continued)

Parameter	Estimate	95% CI
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
All-cause mortality within 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
All-cause mortality within three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Length of emergency department stay		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Length of hospital stay		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Intensive care unit (ICU) admission		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Length of intensive care unit stay		

(continued)

Parameter	Estimate	95% CI
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Return to work at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Return to work at three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Adherence to ATLS® principles		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Quality of life within seven days of discharge		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Quality of life at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Quality of life at three months		
Cluster variance		
Cluster-by-period variance		

(continued)

Parameter	Estimate	95% CI
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Disability within seven days of discharge		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Disability at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		
Disability at three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (rho)		

Abbreviation: CI = Confidence Interval

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